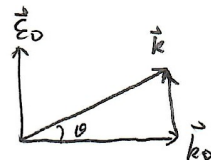


10.17 (a) To apply Eq. (10.132), we need to calculate $\vec{n}_r$. For the geometry

shown on the right, we know

$$\vec{k}_0 = k(0, 0, 1), \quad \vec{k} = k(\sin\theta, 0, \cos\theta).$$



Then, $\vec{k} - \vec{k}_0 = k(\sin\theta, 0, \cos\theta - 1) = 2k\sin\frac{\theta}{2}(\cos\frac{\theta}{2}, 0, -\sin\frac{\theta}{2})$, and therefore, $\vec{n}_r = (\cos\frac{\theta}{2}, 0, -\sin\frac{\theta}{2})$.

For scattering cross section in the E plane, $\vec{E} = (\cos\theta, 0, -\sin\theta)$, $\vec{E}_0 = (1, 0, 0)$, and

$$\begin{aligned} \vec{E}^* \cdot \vec{E}_0 &= \cos\theta, & \vec{E}^* \cdot \vec{E}_r &= -\vec{E}^* \cdot \vec{E}_0 + 2(\vec{n}_r \cdot \vec{E}_0)(\vec{E}^* \cdot \vec{n}_r) \\ & & &= -\cos\theta + 2 \cdot \cos\frac{\theta}{2} \cdot \cos\frac{\theta}{2} = 1. \end{aligned}$$

$$\begin{aligned} \text{Then, } \vec{E}^* \cdot \vec{F} &= \vec{E}^* \cdot \vec{F}_{sh} + \vec{E}^* \cdot \vec{F}_{in} = ika^2 E_0 \cos\theta \frac{J_1(ka\sin\theta)}{ka\sin\theta} + E_0 \frac{a}{2} \exp\{-2ika\sin\frac{\theta}{2}\} \\ &= E_0 \frac{a}{2} \left\{ \cos(2ka\sin\frac{\theta}{2}) + i \left(2\cot\theta J_1(ka\sin\theta) - \sin(2ka\sin\frac{\theta}{2}) \right) \right\}. \end{aligned}$$

$$\text{and } \frac{d\sigma}{dn} = \frac{1}{E_0^2} |\vec{E}^* \cdot \vec{F}|^2 = \frac{a^2}{4} \left[4\cot^2\theta J_1^2(ka\sin\theta) - 4\cot\theta J_1(ka\sin\theta) \sin(2ka\sin\frac{\theta}{2}) + 1 \right].$$

In the H plane, $\vec{E} = \vec{E}_0 = (0, 1, 0)$, and

$$\vec{E}^* \cdot \vec{E}_0 = 1, \quad \vec{E}^* \cdot \vec{E}_r = -1.$$

$$\begin{aligned} \text{Then, } \vec{E}^* \cdot \vec{F} &= \vec{E}^* \cdot \vec{F}_{sh} + \vec{E}^* \cdot \vec{F}_{in} = ika^2 E_0 \frac{J_1(ka\sin\theta)}{ka\sin\theta} - E_0 \frac{a}{2} \exp\{-2ika\sin\frac{\theta}{2}\} \\ &= E_0 \frac{a}{2} \left\{ -\cos(2ka\sin\frac{\theta}{2}) + i \left(2\operatorname{cosec}\theta J_1(ka\sin\theta) + \sin(2ka\sin\frac{\theta}{2}) \right) \right\} \end{aligned}$$

$$\text{and } \frac{d\sigma}{dn} = \frac{1}{E_0^2} |\vec{E}^* \cdot \vec{F}|^2 = \frac{a^2}{4} \left[4\operatorname{cosec}^2\theta J_1^2(ka\sin\theta) + 4\operatorname{cosec}\theta J_1(ka\sin\theta) \sin(2ka\sin\frac{\theta}{2}) + 1 \right].$$